Tae Hwan Jung

AI Engineer · Web3 Engineer

nlkey2022@gmail.com | github.com/graykode | Google Scholar | graykode.github.io

SUMMARY

어떤 도메인이든 빠른 learning curve로 그 분야의 탑 티어가 될 수 있습니다.

"AI knows everything" 시대에 단순히 "정보를 많이 아는 것"은 더 이상 중요하지 않다고 생각합니다. 특히 연구 조직이 아닌, 프로덕트가 더 중요한 스타트업에서 풀어야 하는 문제는 대부분 본인에게 익숙한 도메인에서 시작되지 않습니다. 따라서 저는 본인이 잘하는 문제를 푸는 것보다, **시장에서 꼭 풀어야 하는 문제를 능동적으로 찾아 푸는 방향성**이 앞으로의 엔지니어에게 필요하다고 생각합니다. 이는 문제에 접근하는 방식, 유지보수를 고려한 아키텍처 설계, 그리고 프로덕트를 매니징하는 전 과정을 아우르는 역량입니다.

저는 AI 엔지니어로 커리어를 시작했습니다. 이후 블록체인 필드에서 풀고 싶은 문제를 발견했고, AI에서 했던 것과는 전혀 다른 분야를 zero base부터 시작했습니다. 4년 동안 훌륭한 팀원들과 함께 3–4개월 주기로 프로덕트를 출시하며 6–7번의 프로덕트 사이클을 경험했습니다. 이 과정에서 엔지니어로서는 방향성은 맞지만 아직 개척되지 않은 문제들을 깊이 고민했고, 만든 프로덕트를 시장에서 성공시키기 위한 다양한 시도를 팀원들과 함께 해보았습니다.

창업을 경험하기 이전과 이후의 저는 비교할 수 없을 만큼 달라졌습니다. 엔지니어로서의 주체성뿐 아니라, 문제를 풀어가는 과정에서의 복잡한 인간관계와 낙오의 경험을 통해 훨씬 깊게 성장할 수 있었습니다. 이 과정에서 가장 깊이 배운 덕목은 **윤리성과 책임감**입니다.

제가 풀고자 하는 문제가 있는 곳이라면, 다음 커리어의 도메인은 상관없습니다. 하지만 엔지니어로서는 어떤 분야에서든 AI조차 풀지 못하는 어려운 문제를 프로덕트 관점에서 풀어나가고 싶고, 팀원으로서는 팀에 큰 위기가 닥쳤을 때 책임감 있게 그 무게를 질 수 있는 사람이 되고 싶습니다.

SKILLS

AI Skills

- LLM Distributed Pre-training
- Knowledge Distillation & Compression
- Programming NLP
- Document AI (LayoutLM, OCR)
- PyTorch / Python

Blockchain Skills

- Financial Modeling & DeFi Protocol Design
- Solidity Development & Gas Optimization
- Web3 Full-stack (FE / BE)
- On-chain Data Indexing Pipeline
- Data-driven Market Making & Arbitrage
- TypeScript (main) / Rust

EDUCATION

B.S. in Computer Engineering
2015 – 2022

Kyung Hee University (incl. two years of military service)
Republic of Korea

PUBLICATIONS

Large Product Key Memory for Pretrained Language Models
Gyuwan Kim*, **Tae Hwan Jung*** (*equal contribution)
Findings of EMNLP 2020 · PDF

CommitBERT: Commit Message Generation Using Pre-Trained Programming Language Model
Tae Hwan Jung (sole author)
ACL NLP4Prog Workshop 2021 · PDF · Code

EXPERIENCE

Forward Deployed AI Engineer
Jul 2026 – Present

DSRV
Seoul, Republic of Korea

- Researching x402-based payments for AI agents.
- Building on Canton and security token offering (STO) platforms.

Founding Engineer, Core Protocol
Apr 2022 – May 2026

Clober (Merekat)
Gangnam-gu, Seoul

Built on-chain order-book DEX infrastructure for market makers and trading firms, taking it from zero to one over four years across protocol internals, backend, and frontend alongside co-founders Kevin Hyeon Seok Son and Jihyeok Choy.

Beyond the engineering itself, what four years of founding-stage work at Clober gave me: hands-on zero-to-one product experience (planning, UI/UX, execution, launch); comprehensive domain knowledge in finance; and practical experience in DeFi architecture design and financial modeling. These are the lessons I carry forward.

Launched projects:

Clober v1 — full-stack DEX 2022.04 – 2023.03 | clober.io

- Built Clober v1 across the full stack (smart contracts, backend, and frontend).
- On the contract side, implemented gas-optimized on-chain order matching using segmented segment trees and octopus heaps.
- Write-up: ethresear.ch post · recap thread · gas benchmarks

DEX aggregator on Polygon zkEVM 2023.03 – 2023.06

- Independently built a Rust DEX aggregator for Polygon zkEVM to support Clober's growth on the chain.

Coupon Finance — fixed-rate DeFi lending 2023.06 – 2024.06 | coupon.finance

- Built the frontend and backend of Coupon Finance, a sister protocol built to drive Clober's growth.
- Coupon Finance is a fixed-rate DeFi lending protocol that tokenizes interest rates as tradable "coupons" on an on-chain order book — collapsing the deposit–borrow APY spread down to the order book's bid–ask spread, so both lenders and borrowers get tighter rates than on traditional money markets.

ETH/USDC vault strategy 2024.06 – 2026.05

- Developed and managed Clober's ETH/USDC vault strategy end-to-end, scaling TVL to $500K and closing with a ~6% cumulative return.
- Improved market making with Markov modeling and optimized DEX ↔ Clober arbitrage to cut market-maker losses.
- Built backtesting/simulation pipelines and real-time on-chain monitoring (Ponder-style indexing → Grafana).
- Launched on Monad mainnet (2025.11) and Base (2026.01).

AI Research Engineer

Jul 2021 – Apr 2022

Upstage AI

Yongin, Republic of Korea

Built core Document AI models as an early team member; they later became part of Upstage's enterprise document-understanding product line.

LayoutLM training and fine-tuning

- Trained and fine-tuned LayoutLM — a multimodal transformer that jointly models page layout and text — for Korean document understanding.

OCR labeling dataset

- Built the labeling dataset used to train the document understanding models.

Document parser

- Built a parser for document processing.

AI Engineer

Sep 2020 – Jun 2021

Brunel AI

Mapo-gu, Republic of Korea

Trained and shipped patent-search models into a production AI search product and iterated on real user feedback. This taught me to think about AI from a product perspective.

- Helped scope research problems into AI tasks that could become product features (e.g., deciding which data to collect for long-term value, and framing user problems as concrete ML tasks).
- Improved search quality by developing a model that recommends relevant patents from a user's input query.
- Built the supporting infrastructure for the data pipeline and model serving.

AI Research Engineer (Internship)

Dec 2019 – Jun 2020

NAVER Clova AI

Seongnam, Republic of Korea

Recruited by Jung-Woo Ha to join the LaRva team. Trained Korean language models at scale on distributed GPU clusters and contributed to several research tracks within six months.

Pretraining BERT / RoBERTa on distributed GPU cluster

- Trained on up to 64 V100 GPUs with large-scale corpora in a distributed setup.
- Ran hundreds of pre-training experiments and learned how to avoid CPU/GPU bottlenecks, use mixed precision effectively, and manage large training datasets.

Efficient and lightweight pretrained language models

- Studied how to apply Lample's Product Key Memory (PKM) to pretrained LMs while avoiding catastrophic drift during Masked-LM pretraining and downstream fine-tuning. The work was accepted to Findings of EMNLP 2020.
- Applied knowledge distillation to build a lightweight Korean RoBERTa.

KorQuAD 2.0 leaderboard — 1st place (F1/EM 83.54 / 66.95)

- KorQuAD 2.0 is a reading-comprehension dataset in the spirit of Google's Natural Questions: the model has to find the correct answer span inside a full Wikipedia page, where the answer can be a table, a list, or a paragraph — much harder than single-paragraph QA.

AI Engineer (Internship)

Dec 2018 – Feb 2019

Platfarm

Mapo-gu, Republic of Korea

Built an emoji recommendation model from chat text — my first industry role, where I learned how an engineering team works day to day.

- Annotated and cleaned a raw chat corpus for training.
- Built a sentiment-analysis-based emoji recommendation model.

EXTRACURRICULAR ACTIVITY

KITRI Best of the Best 5th (Vulnerability Track)

Jun 2016 – Dec 2016

Hacking & Security Talent Training Program

Seoul, Republic of Korea

- Selected for the national security talent program; completed the vulnerability analysis track.

Software MAESTRO 10th

Mar 2019 – Nov 2019

Software Developer Talent Training Program

Seoul, Republic of Korea

- Built a federated learning system with on-device TensorFlow training and a Kubernetes-based distributed training pipeline (kubeFL).

SELECTED PROJECTS

nlp-tutorial	— Natural language processing tutorial for deep learning researchers	★ 14,882
abtop	— htop-like real-time monitor for AI coding agents (Claude Code, Codex CLI) in Rust	★ 3,842
nlp-roadmap	— Mind map and keyword roadmap for NLP learners	★ 3,264
distribution-is-all-you-need	— Tutorial on basic probability distributions in deep learning	★ 1,635
gpt-2-Pytorch	— Minimal PyTorch implementation of OpenAI's GPT-2	★ 1,013
xlnet-Pytorch	— Simple XLNet implementation with a PyTorch wrapper	★ 581
commit-autosuggestions	— AI that automatically recommends commit messages (CommitBERT)	★ 389
ai-docstring	— VS Code extension — automatic Python docstring generation via NLP	★ 313
matorage	— Object storage manager for tensors (PyTorch, TensorFlow, Keras)	★ 73
clober-dex/core	— On-chain central limit order book engine (Solidity)	★ 11
clober-dex/v2-sdk	— TypeScript SDK for Clober DEX v2	★ 9